SERIAL COMMUNICATIONS GUIDE FOR THE CP210X

Relevant Devices

This application note applies to the following devices:
CP2101, CP2102, CP2103, CP2104, CP2105, CP2108

1. Introduction

This document is intended for developers creating products based on the CP210x USB to UART Bridge Controller. It provides information about serial communications and how to obtain the port number for a specific CP210x device. Code samples are provided for opening, closing, configuring, reading, and writing to a COM port. Also included is a *GetPortNum* function that can be copied and used to determine the port number on a CP210x device by using its Vendor ID (VID), Product ID (PID), and serial number.

2. Opening a COM Port

Before configuring and using a COM port to send and receive data, it must first be opened. When a COM port is opened, a handle is returned by the *CreateFile()* function that is used from then on for all communication. Here is example code that opens COM3:

```
HANDLE hMasterCOM = CreateFile("\\\\.\\COM3",
              GENERIC_READ | GENERIC_WRITE,
              0,
              0,
              OPEN_EXISTING,
              FILE_ATTRIBUTE_NORMAL | FILE_FLAG_OVERLAPPED,
              0);
```

The first parameter in the *CreateFile()* function is a string that contains the COM port number to use. This string will always be of the form "\\\\.\\COMX" where X is the COM port number to use. The second parameter contains flags describing access, which will be *GENERIC_READ* and *GENERIC_WRITE* for the example in this document, and allows both read and write access. Parameters three and four must always be 0, and the flag in parameter five must always be *OPEN_EXISTING* when using *CreateFile()* for COM applications. The sixth parameter should always contain the *FILE_ATTRIBUTE_NORMAL* flag. In addition, the *FILE_FLAG_OVERLAPPED* is an optional flag that is used when working with asynchronous transfers (this option is used for the example in this document). If overlapped mode is used, functions that read and write to the COM port must specify an OVERLAPPED structure identifying the file pointer, which is demonstrated in section 3.1. and section 3.2. (more information on overlapped I/O is located at http://msdn.microsoft.com/library/en-us/dnfiles/html/msdn_serial.asp?frame=true - serial_topic4). The seventh, and last, parameter must always be 0.

If this function returns successfully, then a handle to the COM port will be assigned to the HANDLE variable. If the function fails, then *INVALID_HANDLE_VALUE* will be returned. Upon return check the handle and if it's valid, then prepare the COM port for data transmission.

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3. Preparing an Open COM Port for Data Transmission

Once a handle is successfully assigned to a COM port several steps must be taken to set it up. The COM port must first be purged and its initial state should be retrieved. Then the COM port's new settings can be assigned and set up by a device control block (DCB) structure (more information is provided on the DCB structure in section 3.3. and at http://msdn.microsoft.com/library/default.asp?url=/library/en-us/devio/base/dcb_str.asp).

3.1. Purging the COM Port

First the COM port should be purged to clear any existing data going to or from the COM port using the *PurgeComm()* function:

```
PurgeComm(hMasterCOM, PURGE_TXABORT | PURGE_RXABORT | PURGE_TXCLEAR | PURGE_RXCLEAR);
```

The first parameter in the *PurgeComm()* function is a handle to the open COM port that will be purged. The second parameter contains flags that further describe what actions should be taken. All four flags, *PURGE_TXABORT*, *PURGE_RXABORT, PURGE_TXCLEAR,* and *PURGE_RXCLEAR* should always be used. The first two flags terminate overlapped write and read operations, and the last two flags clear the output and input buffers.

If this function returns successfully then a non-zero value is returned. If the function fails, then it returns 0. Upon return, check the return value; if it is non-zero, continue to set up the COM port (more information on the *PurgeComm()* function is located at http://msdn.microsoft.com/library/default.asp?url=/library/en-us/devio/base/purgecomm.asp).

3.2. Saving the COM Port's Original State

Since the COM port settings can be modified to meet different needs, it is good practice to obtain the COM port's current state and store it so that when the COM port is closed, the COM port can be restored back to its original state. This can be done using the *GetCommState()* function:

```
DCB dcbMasterInitState;

GetCommState(hMasterCOM, &dcbMasterInitState);
```

The first parameter in the *GetCommState()* function is a handle to the open COM port to obtain settings from. The second parameter is an address to a DCB structure to store the COM port's settings. This DCB structure should also be used as the initial state when specifying new settings for the COM port (see section 3.3.).

If this function returns successfully then a non-zero value is returned. If the function fails, then it returns 0. Upon return, check the return value; if it is non-zero, continue to set up the COM port (more information on the *GetCommState()* function is located at http://msdn.microsoft.com/library/default.asp?url=/library/en-us/devio/base/getcommstate.asp).

3.3. Setting up a DCB Structure to Set the New COM State

All of a COM port's settings are stored in a DCB structure. In section 3.2. a DCB structure was retrieved that contained the initial settings of the COM port by using the *GetCommState()* function. To change a COM port's settings, a DCB structure must be created and filled out with the desired settings. Then the *SetCommState()* function can be used to activate those settings:

```
DCB dcbMaster = dcbMasterInitState;

dcbMaster.BaudRate= 57600;
dcbMaster.Parity= NOPARITY;
dcbMaster.ByteSize= 8;
dcbMaster.StopBits= ONESTOPBIT;

SetCommState(hMasterCOM, &dcbMaster);

Delay(60);
```

Here a new DCB structure dcbMaster has been initialized to dcbMasterInitState, which are the current settings of the COM port. After it has been initialized to the current settings, new settings can be assigned.

3.3.1. Baud Rate

The baud rate property is set to 57600 bps, but can be set to any of the baud rates supported by the CP210x. (See the current data sheet for the list of supported baud rates for the CP210x.)

3.3.2. Parity

The parity is set to NOPARITY, however it can also be set to ODDPARITY, EVENPARITY, SPACEPARITY, and MARKPARITY if supported by the CP210x. (See the current data sheet for the list of supported parities for the CP210x.)

3.3.3. Byte Size

The byte size is set to 8, so there are 8 data bits in every byte of data sent. This can also be set to 5, 6, or 7 if supported by the CP210x. (see the data sheet for the list of supported byte sizes for the CP210x.)

3.3.4. Stop Bits

The stop bits are set to ONESTOPBIT, but could also be set to TWOSTOPBITS or ONE5STOPBITS (1.5). (See the current data sheet for the list of supported stop bits for the CP210x.) All combinations of data and stop bits can be used except for the combination of 5 data bits with 2 stop bits and the combination of 6, 7, or 8 data bits with 1.5 stop bits.

After each of these settings is set to the desired value, the *SetCommState()* function can be called to set up the COM port. The first parameter in the *SetCommState()* function is a handle to the open COM port to change the settings on. The second parameter is an address to a DCB structure containing the COM port's new settings 2 (more information on serial settings using DCB structures is located at http://msdn.microsoft.com/library/en-us/dnfiles/html/msdn_serial.asp?frame=true - serial_topic6).

If this function returns successfully, a non-zero value is returned. If the function fails, it returns 0. Upon return, check the return value; if it is non-zero, delay for 60 ms to allow time for the settings to change and then continue to set up the COM port. This delay is not required; however, a conservative time of 60 ms is good practice to ensure that the settings are changed before any other operations take place.

4. Transmitting Data Across the COM Port

Once the COM port is successfully opened and configured, data can be written or read.

4.1. Writing Data

There are several things that need to happen in a write, so it is a good idea to create a function for the writes to be called whenever a write must occur. Here is an example of a write function:

```
bool WriteData(HANDLE handle, BYTE* data, DWORD length, DWORD* dwWritten)
{
   bool success= false;
   OVERLAPPED o= {0};

   o.hEvent = CreateEvent(NULL, FALSE, FALSE, NULL);

   if (!WriteFile(handle, (LPCVOID)data, length, dwWritten, &o))
   {
      if (GetLastError() == ERROR_IO_PENDING)
         if (WaitForSingleObject(o.hEvent, INFINITE) == WAIT_OBJECT_0)
            if (GetOverlappedResult(handle, &o, dwWritten, FALSE))
            success = true;
   }
   else
      success = true;

   if (*dwWritten != length)
      success = false;

   CloseHandle(o.hEvent);

   return success;
}
```

The parameters passed in to this function are the handle to an open COM port, a pointer to an array of bytes that will be written, the number if bytes that are in the array, and a pointer to a variable to store and return the number of bytes written.

Two local variables are declared at the beginning of the function: a bool named success that will store the success of the write (this is initialized to false, and only set true when the write succeeds) and an overlapped object o which is passed to the *WriteFile()* function and alerts if the transfer is complete or not (this is always initialized to {0} before the hEvent is assigned). Creating an event with the CreateEvent(NULL, FALSE, FALSE, NULL) function sets the hEvent property of o to prepare it to be passed to the *WriteFile()* function (more information on *CreateEvent()* is located at http://msdn.microsoft.com/library/default.asp?url=/library/en-us/dllproc/base/createevent.asp).

Next, the *WriteFile()* function is called with the handle, data, length of the data, and variable to store the amount of data that was written (more information on *WriteFile()* is located at http://msdn.microsoft.com/library/default.asp?url=/library/en-us/fileio/base/writefile.asp). If this function returns successfully, a non-zero value is returned. If the function fails, it returns 0. The if statement will determine if the write succeeded and if it did not, the last error is retrieved to see if there really was an error or the write just wasn't finished. If *ERROR_IO_PENDING* is returned then object o is then waited on until either the write finishes or fails (if something other than *ERROR_IO_PENDING* is returned by the *GetLastError()* function, then there is the possibility of surprise removal; see “8. Application Design Notes” for comments on surprise removal). When the wait is over, the result is obtained so that the amount of bytes written is updated. The success variable will then be assigned with the appropriate value, and the handle of o.hEvent is closed. Then the amount of bytes written is checked, and finally the function returns the success of the write, which will be true if the write successfully completed.

4.2. Reading Data

There are several things that need to happen in a read, so it is a good idea to create a function for the reads to be called whenever a read must occur. Here is an example of a read function:

```
bool ReadData(HANDLE handle, BYTE* data, DWORD length, DWORD* dwRead, UINT timeout)
{
   bool success= false;
   OVERLAPPEDo= {0};

   o.hEvent = CreateEvent(NULL, FALSE, FALSE, NULL);

   if (!ReadFile(handle, data, length, dwRead, &o))
{
      if (GetLastError() == ERROR_IO_PENDING)
         if (WaitForSingleObject(o.hEvent, timeout) == WAIT_OBJECT_0)
            success = true;
         GetOverlappedResult(handle, &o, dwRead, FALSE);
   }
   else
      success = true;

   CloseHandle(o.hEvent);

   return success;
}
```

The parameters passed in to this function are the handle to an open COM port, a pointer to an array of bytes that will be read, the number if bytes that are in the array, a pointer to a variable to store and return the number of bytes read, and a timeout value.

Two local variables are declared at the beginning of the function: a bool named success that will store the success of the read (this is initialized to false, and only set true when the read succeeds), and an overlapped object o which is passed to the *ReadFile()* function and alerts if the transfer is complete or not (this is always initialized to {0} before the hEvent is assigned). Creating an event with the CreateEvent(NULL, FALSE, FALSE, NULL) function sets the hEvent property of o to prepare it to be passed to the *ReadFile()* function (more information on *CreateEvent()* is located at http://msdn.microsoft.com/library/default.asp?url=/library/en-us/dllproc/base/createevent.asp).

Next, the *ReadFile()* function is called with the handle, data, length of the data, and variable to store the amount of data that was written (more information on the *ReadFile()* function is located at http://msdn.microsoft.com/library/default.asp?url=/library/en-us/fileio/base/readfile.asp). If this function returns successfully then a non-zero value is returned. If the function fails, then it will return 0. The if statement will determine if the write succeeded and if it didn't, the last error is retrieved to see if there really was an error or the write just wasn't finished. If *ERROR_IO_PENDING* is returned then object o is then waited on until either the write finishes or fails (if something other than *ERROR_IO_PENDING* is returned by the *GetLastError()* function, then there is the possibility of surprise removal; see section 8. "Application Design Notes" for comments on surprise removal). When the wait is over, the result is obtained so that the amount of bytes read is updated. The success variable will then be assigned with the appropriate value, and the handle of o.hEvent is closed. Finally, the function returns the success of the read, which will be true if the read successfully completed.

5. Closing the COM Port

After all communication is finished, then the COM port should then be closed. First, the COM port should be set back to its initial state, and then the handle to the COM port should be closed and set to an invalid handle. Example code is shown below:

```
SetCommState(hMasterCOM, &dcbMasterInitState);

Delay(60);

CloseHandle(hMasterCOM);
hMasterCOM = INVALID_HANDLE_VALUE;
```

The *SetCommState()* function works the same as described in "3.3. Setting up a DCB Structure to Set the New COM State" on page 2. A delay of 60 ms is used to make sure the settings have time to be set. Finally the device is closed using the *CloseHandle()* function. This function just takes in the handle of the COM port. After this function is called, it is important to set the variable to an *INVALID_HANDLE_VALUE*.

6. Sample program to Demonstrate Serial Communications

Included in the AN197SW.zip is a directory named "CP210xSerialTest" which contains the source code and executables for a Visual Studio project that makes use of all the serial communication functions described in section 3., section 4., and section 5. The program is a basic dialog based application that accepts two COM port numbers, and then will send a test array of 64 bytes of data back and forth between them.

7. Discovering CP210x COM Port

To use the described functionality with a COM port, the number of the COM port needs to be known. In order to find out the COM port number of a CP210x device, the VID, PID, and serial number are used to lookup a registry key. This key is different between Windows XP/2000/Server 2003/Vista and Windows 98. Here are the keys that will need to be looked up:

WinXP/2000/Server 2003/Vista/7 (Driver Version 5.0 and higher):

HKLM\System\CurrentControlSet\Enum\USB\Vid_xxxx&Pid_yyyy\zzzz\Device Parameters\PortName

WinXP/2000/Server 2003/Vista/7 (Driver Version 4.40 and lower):

HKLM\System\CurrentControlSet\Enum\USB\Vid_xxxx&Pid_yyyy&Mi_00\zzzz_00\Device Parameters\PortName

(where xxxx is the VID, yyyy is the PID, and zzzz is the serial number)

7.1. Windows XP/2000/Server 2003/Vista/7

To find the port number in Windows XP/2000/Server 2003/Vista/7, the corresponding key listed above needs to be opened. This is done using several registry calls using Windows API functions. The function is passed the VID, PID, and serial number, and the return value is either the port number that the CP210x is located on, or a –1 if there is a failure. The function below will find the CP210x port number in Windows XP/2000/Server 2003/Vista/7 (this function can be copied straight into application code that needs to discover the COM port number):

```
int GetPortNum(WORD vid, WORD pid, char* ser, BYTE ifc)
{
   //Variables used for Registry access
   HKEY tmpKey, tmpSubKey, tmpPortKey;
   CString portKeyString;
   DWORD valtype;
   wchar_t* portString;
   wchar_t* parentIdString;
   DWORD length = 100;
   portString = new wchar_t[101];
   parentIdString = new wchar_t[101];
   wchar_t tmpPortString[10];

   //Set portnum to -1, so if there is an error we will
   //know by returning a negative port value
   int portNum = -1;

   //  Open keys to get to the key where the port number is located. This key is:
   if (ERROR_SUCCESS == RegOpenKeyEx(HKEY_LOCAL_MACHINE, L"SYSTEM\\CurrentControlSet\\", 0,
KEY_READ, &tmpKey))
   {
      if (ERROR_SUCCESS == RegOpenKey(tmpKey, L"Enum\\USB\\", &tmpSubKey))
      {
         //Loop through and replace spaces for WinXP2000Vista
         int i = 0;
         while (ser[i] != '\0')
         {
            if (ser[i] == 0x20)
               ser[i] = '_';
            i++;
         }

         //The portkey string should look like this
         //"Vid_XXXX&Pid_XXXX\\XXXX" where the XXXX's are Vid, Pid and serial string
         portKeyString = L"Vid_";
         _itow_s((int)vid, tmpPortString, 10, 16);
         portKeyString += tmpPortString;
         portKeyString += L"&Pid_";
         _itow_s((int)pid, tmpPortString, 10, 16);
         portKeyString += tmpPortString;
         portKeyString += L"\\";
         portKeyString += ser;
         portKeyString += L"\\Device Parameters\\";

         //If the portkey string is in the registry, then go ahead and open the portname
         if (ERROR_SUCCESS == RegOpenKeyEx(tmpSubKey, portKeyString, 0, KEY_READ,
&tmpPortKey))
         {
            // CP2101/2/3/4
            // This will only work for single interface devices, if it fails
```

```
            // we then look to see if it is a multi interface device
            if (ERROR_SUCCESS == RegQueryValueEx(tmpPortKey, L"PortName", NULL, &valtype,
(unsigned char *)portString, &length))
            {
               // When we obtain this key, it will be in string format of
               // "COMXX" where XX is the port. Simply make the first three
               // elements of the string 0, and call the atoi function to obtain
               // the number of the port.
               portString[0] = '0';
               portString[1] = '0';
               portString[2] = '0';
               portNum = _wtoi(portString);
            }
            else
            {
               //Close the original port key
               RegCloseKey(tmpPortKey);

               // Reset our port key string to not include device paramaters
               portKeyString = L"Vid_";
               _itow_s((int)vid, tmpPortString, 10, 16);
               portKeyString += tmpPortString;
               portKeyString += L"&Pid_";
               _itow_s((int)pid, tmpPortString, 10, 16);
               portKeyString += tmpPortString;
               portKeyString += L"\\";
               portKeyString += ser;

               //If the portkey string is in the registry, then get the parent ID to start
looking at
               //the multi interface devices
               if (ERROR_SUCCESS == RegOpenKeyEx(tmpSubKey, portKeyString, 0, KEY_READ,
&tmpPortKey))
               {
               if (ERROR_SUCCESS == RegQueryValueEx(tmpPortKey, L"ParentIdPrefix", NULL,
&valtype, (unsigned char *)parentIdString, &length))
               {
               //Close the original port key, reopen the multi interface
               //devices
               RegCloseKey(tmpPortKey);

               // Reset our port key string to not include device paramaters and
               // the parent ID string, as well as interface info
               portKeyString = L"Vid_";
               _itow_s((int)vid, tmpPortString, 10, 16);
               portKeyString += tmpPortString;
               portKeyString += L"&Pid_";
               _itow_s((int)pid, tmpPortString, 10, 16);
               portKeyString += tmpPortString;
               portKeyString += L"&MI_";
               _itow_s((int)ifc, tmpPortString, 10, 16);
               if (ifc <= 0x0F)
               {
               portKeyString += L"0";
               }
               portKeyString += tmpPortString;
               portKeyString += L"\\";
               portKeyString += parentIdString;
               portKeyString += L"&00";
```

```
                if (ifc <= 0x0F)
                {
                portKeyString += L"0";
                }
                portKeyString += tmpPortString;
                portKeyString += "\\Device Parameters\\";

                if  (ERROR_SUCCESS  ==  RegOpenKeyEx(tmpSubKey,  portKeyString,  0,  KEY_READ,
&tmpPortKey))
                {
                if (ERROR_SUCCESS == RegQueryValueEx(tmpPortKey, L"PortName", NULL, &valtype,
(unsigned char *)portString, &length))
                {
                // When we obtain this key, it will be in string format of
                // "COMXX" where XX is the port. Simply make the first three
                // elements of the string 0, and call the atoi function to obtain
                // the number of the port.
                portString[0] = '0';
                portString[1] = '0';
                portString[2] = '0';
                portNum = _wtoi(portString);
                }
                }
                }
                }
            }
            //Make sure to close all open keys for cleanup
            RegCloseKey(tmpPortKey);
        }

        RegCloseKey(tmpSubKey);
    }
    RegCloseKey(tmpKey);
  }
  RegCloseKey(HKEY_LOCAL_MACHINE);

  delete portString;
  delete parentIdString;

  // Return the number of the port the device is connected too
  return portNum;
}
```

7.2. Sample Program to Retrieve a CP210x COM Port

Included in the AN197SW.zip is a directory named "CP210xPortNumExample" that contains the source code and executables for a Visual Studio project that makes use of these functions. In the program, the number of CP210x devices are found and then for each one a port number is retrieved for it and listed in the edit box. (This program also makes use of the *CP210xManufacturing.DLL* to find the VID, PID, and serial number on a device. The use of this DLL is defined in more detail in "AN144: CP21xx Device Customization Guide".)

8. Application Design Notes

The functions used in section 3., section 4., and section 5. are Windows COMM API functions. The examples provided are just samples of the recommended way of dealing with serial communication. For more specific information on these functions, see the MSDN website at: http://msdn.microsoft.com/library/default.asp.

It should also be noted that the *SetCommState()* function does not save the settings between opening and closing the COM port. As stated before, it is good practice to get the current settings after the COM port is opened, and then restore them before it is closed.

All of the functions here will return an error code. It is a good idea to nest these functions in order to catch errors if they occur by using the *GetLastError()* function. This will also solve any surprise removal problems by allowing the discovery of an invalid handle to be found and dealt with. The example application (CP210xSerialTest) has several cases that will detect surprise removal. In this example, there are checks on every function to make sure that the return code is true. If it is not, then it will display where the error occurred in the output window. As long as correct and supported settings are passed to the functions they should execute normally. Most failures can occur from having an *INVALID_HANDLE_VALUE*, however, the handles must be set to this value after a surprise removal occurs.

Because regular COM ports will always be visible, then data can always be written to them successfully, even if there is no way to read it. However, because the CP210x is a virtual COM port, if the device is removed, then the handle that it uses becomes invalid when trying to write to it. If for some reason the CP210x device is unplugged the write will fail and *ERROR_OPERATION_ABORTED* will be returned by *GetLastError()*. When this happens, the handle needs to be closed and then set to *INVALID_HANDLE_VALUE*. Alternatively, a regular COM port can always be read from, but if there is no data then it will time out. When using the CP210x as the virtual COM port and it is removed before a read occurs, then the read will fail and *ERROR_ACCESS_DENIED* will be returned by *GetLastError()*. Again when this happens, the handle needs to be closed and then set to *INVALID_HANDLE_VALUE*.

9. References

MSDN - use this to search for specific Windows API functions

http://msdn.microsoft.com/

Serial Communication Reference

http://msdn.microsoft.com/library/default.asp?url=/library/en-us/dnfiles/html/msdn_serial.asp

Communication Resource Reference

http://msdn.microsoft.com/library/default.asp?url=/library/en-us/devio/base/communications_resources.asp

DOCUMENT CHANGE LIST

Revision 0.3 to Revision 0.4

- Updated "7.2. Sample Program to Retrieve a CP210x COM Port" on page 10.

Revision 0.4 to Revision 0.5

- Added CP2103 to Relevant Devices on page 1.

Revision 0.5 to Revision 0.6

- Updated XP/2000 references to include Server 2003/Vista.
- Updated WinXP/2000 key listing in section "7. Discovering CP210x COM Port" on page 6.
- Updated code in section "7.1. Windows XP/2000/ Server 2003/Vista/7" on page 7.

Revision 0.6 to Revision 0.7

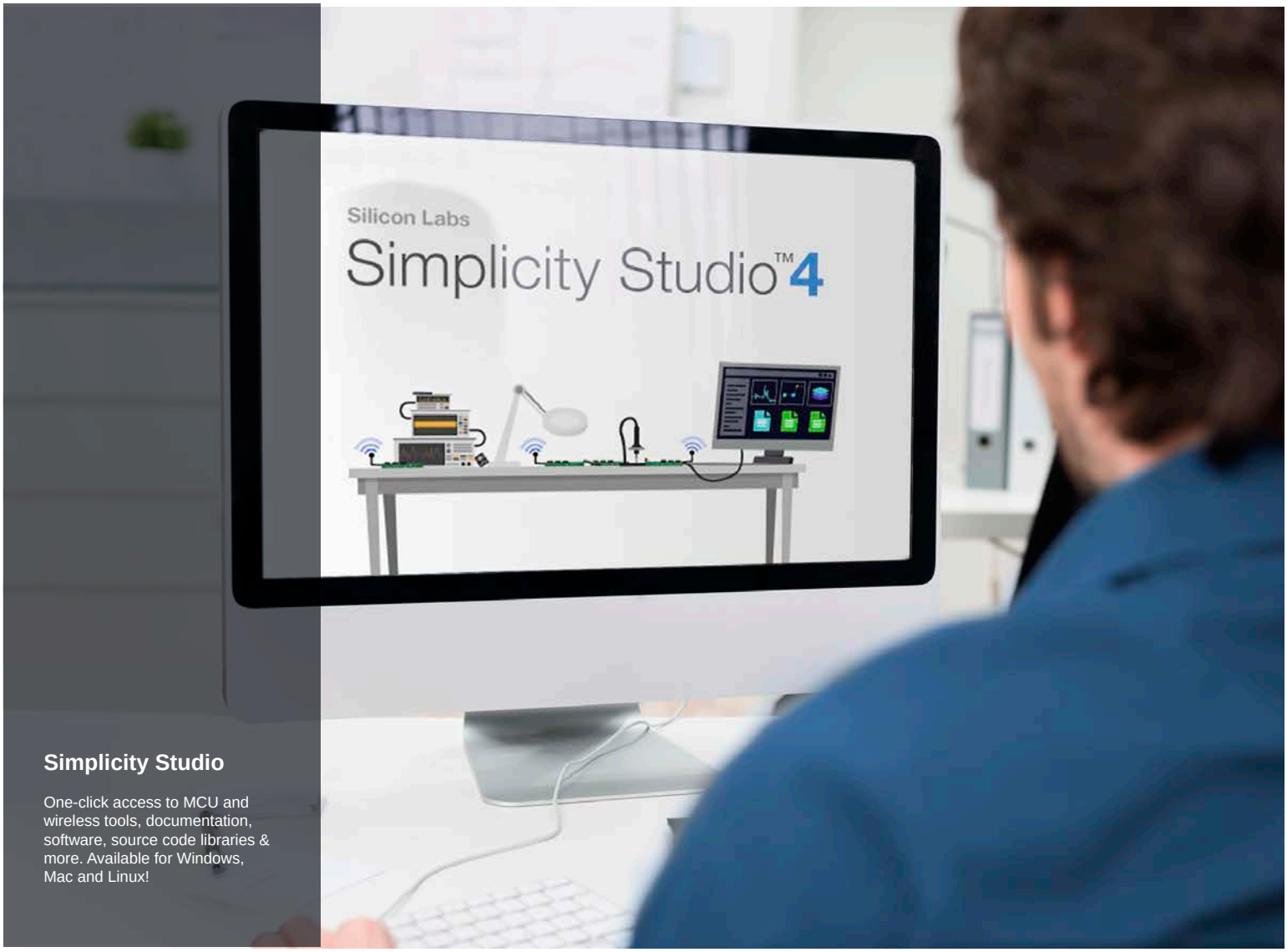
- Corrected registry paths in "7. Discovering CP210x COM Port" on page 6.

Revision 0.7 to Revision 0.8

- Added CP2104/5 to Relevant Devices.
- Updated GetPortNum function and added support for Windows 7 in Section 7.
- Removed support for Windows 98.

Revision 0.8 to Revision 0.9

- Added CP2108 to Relevant Devices list.



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